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1 Predicting Customer Churn in a Telecommunications Business

Kyle Gotzman

DSC630-T302 Predictive Analytics

Bellevue University

Professor Andrew Hua

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1.1 Introduction of Topic/Problem

Customer churn, when subscribers voluntarily cancel their service, represents one of the most critical challenges facing telecommunications companies today. In an industry characterized by high customer acquisition costs and intense competition, retaining existing customers is significantly more cost-effective than acquiring new ones. Research consistently demonstrates that acquiring a new customer can cost five to seven times more than retaining an existing one (Siddiqi, 2012), making proactive churn prevention a high-value business strategy.

The central business question driving this project is: **Can customer demographics, service subscriptions, and billing data be used to predict which customers are likely to cancel service, and what actions should the company take to reduce churn risk?**

This problem is important for several reasons. First, telecommunications companies operate in a saturated market where organic growth is limited, making retention the primary lever for revenue stability. Second, churn is not uniformly distributed; specific customer segments exhibit dramatically higher cancellation rates, creating an opportunity for targeted intervention. Third, predictive models can transform reactive customer service into proactive retention campaigns, contacting at-risk customers before they make a cancellation decision.

The primary stakeholders for this analysis include marketing and retention teams who design intervention campaigns, product managers who bundle service offerings, customer service leadership who allocate support resources, and executive leadership who set strategic priorities around customer lifetime value. A well-performing churn prediction model enables these stakeholders to allocate limited retention budgets to the customers who are most likely to leave, and most likely to respond to outreach.

This project follows a structured predictive analytics workflow: exploratory data analysis to understand churn patterns, data preparation to engineer a modeling-ready dataset, model training using multiple classification algorithms, evaluation against predefined success criteria, and interpretation of results to inform actionable business recommendations.

1.2 Overview of Data Used

1.2.1 Data Source

The dataset used in this project is the IBM Telco Customer Churn dataset, publicly available through the IBM Watson Analytics community and mirrored on Kaggle (IBM, n.d.). It contains 7,043 customer records from a fictional telecommunications company, with each record representing a unique customer and their service history.

1.2.2 Feature Description

The dataset contains 21 columns spanning three feature categories:

Demographics (4 features): - `gender` — Male or Female - `SeniorCitizen` — Whether the customer is a senior citizen (binary) - `Partner` — Whether the customer has a partner - `Dependents` — Whether the customer has dependents

Service Subscriptions (9 features): - `PhoneService` — Whether the customer has phone service - `MultipleLines` — Whether the customer has multiple phone lines - `InternetService` — Type of internet service (DSL, Fiber optic, or None) - `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport` — Add-on services - `StreamingTV`, `StreamingMovies` — Streaming service subscriptions

Account Information (4 features): - `tenure` — Number of months the customer has been with the company (0–72) - `Contract` — Contract type (Month-to-month, One year, Two year) - `PaperlessBilling` — Whether the customer uses paperless billing - `PaymentMethod` — Payment method (Electronic check, Mailed check, Bank transfer, Credit card) - `MonthlyCharges` — Monthly billing amount - `TotalCharges` — Cumulative billing amount

Target Variable: - `Churn` — Whether the customer churned (Yes/No)

1.2.3 Data Quality

The dataset is relatively clean with one notable issue: the `TotalCharges` column is stored as text, and 11 rows contain blank strings instead of numeric values. All 11 correspond to customers with `tenure = 0` (brand-new subscribers who have not completed a billing cycle). These were imputed with `MonthlyCharges` since that represents the only charge these customers would owe, a defensible choice that avoids discarding the newest, most churn-prone customers.

1.2.4 Why This Data Is Useful

This dataset is well-suited for churn prediction because it contains labeled outcomes (churn/no churn), a mix of categorical and numeric features spanning demographics, behavior, and financials, and sufficient volume (7,043 records) to support train/test splitting with meaningful statistical

power. The feature set maps directly to actionable business levers; contract type, add-on services, and payment methods are all elements that retention teams can influence through targeted campaigns.

1.3 Methods of Analysis

1.3.1 Exploratory Data Analysis

The analysis began with comprehensive exploratory data analysis (EDA) to understand the structure of the data and identify patterns associated with churn. Key EDA activities included:

1. **Class distribution analysis** — The target variable exhibits moderate class imbalance: approximately 26.5% of customers churned while 73.5% did not. This imbalance informed the modeling strategy (balanced class weights, stratified splitting).
2. **Univariate analysis** — Each feature was examined individually for its relationship to churn using bar charts (categorical features) and box plots (numeric features).
3. **Bivariate relationships** — Interactions between features were explored, particularly the contract type × internet service interaction which revealed that month-to-month fiber optic customers represent the highest-risk segment.
4. **Correlation analysis** — Numeric features were examined for multicollinearity, revealing the expected strong correlation between tenure and total charges.

1.3.2 Data Preparation

The preprocessing pipeline consisted of the following steps:

1. **Drop `customerID`** — A unique identifier with no predictive value.
2. **Clean `TotalCharges`** — Convert from text to numeric; impute 11 blank values with `MonthlyCharges` for tenure-zero customers.
3. **Encode categorical features** — One-hot encoding with `drop_first=True` to avoid multicollinearity, producing 30 binary features from the original 16 categorical columns.
4. **Train/test split** — 80/20 stratified split to preserve churn class proportions in both sets (training: 5,634 rows; test: 1,409 rows).
5. **Feature scaling** — `StandardScaler` applied to `tenure`, `MonthlyCharges`, and `TotalCharges` for logistic regression convergence.
6. **Class imbalance handling** — `class_weight='balanced'` in applicable models rather than resampling, preserving the natural data distribution.

1.3.3 Model Selection

Four models were trained and evaluated, each chosen for a specific purpose:

Model	Purpose	Configuration
Naive Baseline	Establish performance floor	Always predicts majority class
Logistic Regression	Interpretable coefficients for business insight	`class_weight='balanced'`, `max_iter=1000`

Model	Purpose	Configuration
Decision Tree	Non-linear relationships, visualizable rules	`max_depth=5`, `class_weight='balanced'`
Random Forest	Ensemble accuracy, feature interactions	`n_estimators=200`, `max_depth=10`, `class_weight='balanced'`

1.3.4 Evaluation Metrics

Models were evaluated against predefined success criteria on the held-out test set:

- **Recall 0.70** — Prioritized because missing at-risk customers is costly for retention campaigns
- **AUC-ROC 0.80** — Measures overall discriminative ability across all classification thresholds
- **Precision 0.50** — Ensures retention campaigns are not wasted on customers who would not have churned
- **F1-score 0.60** — Balances precision and recall into a single summary metric

The naive baseline was included to demonstrate why raw accuracy is misleading for imbalanced classification — a model that always predicts "no churn" achieves 73.5% accuracy but 0% recall on the class of interest.

1.4 Results & Findings Explained

1.4.1 Model Performance Comparison

All three trained models met the predefined success criteria, substantially outperforming the naive baseline:

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Naive Baseline	0.7346	0.0000	0.0000	0.0000	0.5000
Logistic Regression	0.7381	0.5043	0.7834	0.6136	0.8418
Decision Tree	0.7346	0.5000	0.8075	0.6176	0.8308
Random Forest	0.7566	0.5281	0.7781	0.6292	0.8409

The **Random Forest** achieved the best overall balance with the highest accuracy (75.7%), precision (52.8%), and F1-score (62.9%). The **Decision Tree** achieved the highest recall (80.8%), identifying the most churners at the cost of slightly more false positives. **Logistic Regression** provided competitive performance with the advantage of fully interpretable coefficients.

The naive baseline confirms that raw accuracy is meaningless for this problem — 73.5% accuracy with zero ability to identify churners demonstrates why recall and AUC-ROC are the appropriate primary metrics.

1.4.2 Key Churn Drivers

Analysis of feature importances (Random Forest) and logistic regression coefficients consistently identified the following top churn predictors:

1. **Contract type** — Month-to-month customers churn at approximately 43%, compared to ~11% for one-year contracts and ~3% for two-year contracts. This is the single most important predictor across all models.
2. **Tenure** — Strongly negatively correlated with churn. Customers in their first 12 months represent the highest-risk window, with churn rates declining steadily as tenure increases.
3. **Internet service type** — Fiber optic customers churn at significantly higher rates than DSL customers, likely driven by greater competition and higher monthly charges in urban/suburban markets where fiber is available.
4. **Add-on services** — Customers without online security, tech support, online backup, or device protection churn at substantially higher rates. Bundling these services may create stickiness.
5. **Payment method** — Electronic check users churn at notably higher rates than customers using automatic payment methods (bank transfer, credit card), suggesting that manual payment friction reduces commitment.
6. **Monthly charges** — Higher monthly bills are associated with increased churn, particularly among month-to-month customers who have no switching cost.

1.4.3 Fairness Analysis

Demographic fairness monitoring revealed:

- **Gender** shows minimal prediction disparity — recall and precision are comparable between male and female subgroups.
- **Senior citizen** status shows a modest gap (seniors have slightly lower recall), warranting continued monitoring but not indicating systematic bias.

1.4.4 Recommended Model for Deployment

The **Random Forest** is recommended as the primary model for the following reasons:

- Highest F1-score (0.6292) — best precision-recall balance
- Competitive AUC-ROC (0.8409) — comparable to logistic regression
- Robust to feature interactions and non-linear relationships
- Feature importance output directly supports business interpretation

However, logistic regression should be maintained as a complementary model because its coefficients provide directional insight (which factors push toward/away from churn) that is valuable for designing specific intervention strategies.

1.5 Conclusion

1.5.1 Summary of Findings

This project demonstrates that customer churn in telecommunications can be effectively predicted using readily available customer, service, and billing data. All three trained classification models met the predefined success criteria, with the Random Forest delivering the strongest overall performance (AUC-ROC = 0.84, Recall = 0.78, F1 = 0.63).

The analysis reveals that churn is not random — it is concentrated among identifiable segments with specific, actionable characteristics. The strongest predictors (contract type, tenure, internet service, and add-on subscriptions) are all factors that the business can influence through retention campaigns.

1.5.2 Business Recommendations

Based on the model results and feature analysis, the following prioritized interventions are recommended:

Priority	Action	Target Segment	Rationale
High	Offer contract upgrade incentives	Month-to-month customers	Contract type is the #1 churn driver
High	Proactive outreach in first 12 months	New customers (tenure $<$ 12 mo)	Early tenure is the highest-risk window
Medium	Bundle add-on services at discount	Customers without security/support	Add-ons strongly associated with retention
Medium	Strengthen competitive positioning	Fiber optic subscribers	Higher churn driven by market competition
Low	Encourage automatic payment enrollment	Electronic check users	Manual payment associated with higher churn

1.5.3 Deployment Readiness

The model is ready for a **pilot deployment** with the following caveats:

- **Strengths:** Meets all predefined performance thresholds, uses standard and interpretable algorithms, requires only data already captured in billing systems.
- **Limitations:** Precision of ~53% means roughly half of flagged customers would not have churned — acceptable for low-cost interventions (email campaigns) but may need improvement for high-cost actions (discounted contracts).
- **Recommended next steps:** Threshold tuning to optimize the precision-recall tradeoff for specific campaign costs, A/B testing of interventions on model-flagged vs. random customers, and periodic retraining as customer behavior evolves.

1.5.4 Future Work

Several improvements could enhance model performance in production:

1. **Feature engineering** — Interaction terms (e.g., tenure $\times$ contract type), rolling behavioral aggregates
2. **Advanced algorithms** — Gradient boosting (XGBoost, LightGBM) for potentially higher predictive power

3. **Hyperparameter tuning** — Systematic grid/random search to optimize model configuration
4. **Temporal validation** — Time-based splitting to simulate real deployment conditions
5. **Cost-sensitive learning** — Incorporating actual retention campaign costs into the optimization objective

1.5.5 Ethical Considerations

Several ethical dimensions must be considered for any production deployment:

- **Demographic fairness:** While gender showed no meaningful prediction disparity, senior citizen status exhibited a modest gap. Periodic fairness audits should be mandatory.
- **Data privacy:** The model uses billing and service data that customers provided for service delivery purposes. Using it for retention targeting should be transparently disclosed in privacy policies.
- **Intervention ethics:** Targeted retention offers (discounts, upgrades) should be applied equitably. If the model disproportionately flags certain demographic groups, differential treatment could constitute discrimination.
- **Transparency:** Customers flagged as at-risk should not be subjected to aggressive or manipulative retention tactics. Ethical retention means improving the genuine value proposition.
- **Model drift:** Customer behavior changes over time. A deployed model that is not regularly retrained may develop emergent biases as the population shifts.

To mitigate these concerns, any production deployment should include: (1) quarterly fairness audits across demographic subgroups, (2) clear privacy disclosures, (3) human oversight of high-impact decisions, and (4) scheduled model retraining with drift monitoring.

1.6 References

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1.7 —

2 Appendix: Python Code

The following sections contain the complete Python implementation for this project, covering data loading, exploratory analysis, preprocessing, model training, evaluation, and visualization.

2.0.1 Setup and Imports

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
from pathlib import Path
import warnings

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.dummy import DummyClassifier
from sklearn.metrics import (classification_report, confusion_matrix,
                             roc_auc_score, roc_curve, accuracy_score)

warnings.filterwarnings('ignore')

sns.set_theme(style='whitegrid', palette='muted')
plt.rcParams['figure.figsize'] = (10, 6)
plt.rcParams['figure.dpi'] = 100
```

2.0.2 Data Loading and Initial Exploration

[2]:
```
DATA_PATH = Path('Telco-Customer-Churn.csv')

df_raw = pd.read_csv(DATA_PATH)
print(f'Shape: {df_raw.shape[0]:,} rows × {df_raw.shape[1]} columns')
df_raw.head(3)
```

```
Shape: 7,043 rows × 21 columns
```

[2]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
0	7590-VHVEG	Female	0	Yes	No	1	No	

```
1  5575-GNVDE    Male              0      No         No      34           Yes
2  3668-QPYBK    Male              0      No         No       2           Yes

      MultipleLines InternetService OnlineSecurity  … DeviceProtection  \
0  No phone service             DSL             No  …                No
1                No             DSL            Yes  …               Yes
2                No             DSL            Yes  …                No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes

      PaymentMethod MonthlyCharges  TotalCharges Churn
0  Electronic check          29.85         29.85    No
1      Mailed check          56.95        1889.5    No
2      Mailed check          53.85        108.15   Yes

[3 rows x 21 columns]
```

[3]:
```
info_df = pd.DataFrame({
    'dtype': [str(df_raw[c].dtype).replace('object', 'str') for c in df_raw.
 ↪columns],
    'non_null': df_raw.count().values,
    'missing': df_raw.isnull().sum().values,
    'unique': df_raw.nunique().values
}, index=df_raw.columns)
info_df
```

[3]:
```
                  dtype  non_null  missing  unique
customerID          str      7043        0    7043
gender              str      7043        0       2
SeniorCitizen     int64      7043        0       2
Partner             str      7043        0       2
Dependents          str      7043        0       2
tenure            int64      7043        0      73
PhoneService        str      7043        0       2
MultipleLines       str      7043        0       3
InternetService     str      7043        0       3
OnlineSecurity      str      7043        0       3
OnlineBackup        str      7043        0       3
DeviceProtection    str      7043        0       3
TechSupport         str      7043        0       3
StreamingTV         str      7043        0       3
StreamingMovies     str      7043        0       3
Contract            str      7043        0       3
PaperlessBilling    str      7043        0       2
```

```
PaymentMethod          str      7043        0       4
MonthlyCharges     float64      7043        0    1585
TotalCharges           str      7043        0    6531
Churn                  str      7043        0       2
```

2.0.3 Data Cleaning

[4]:
```
df = df_raw.copy()

blank_total = (df['TotalCharges'].astype(str).str.strip() == '').sum()
print(f'Blank TotalCharges values: {blank_total}')

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
print(f'TotalCharges missing after conversion: {df["TotalCharges"].isna().
  ↪sum()}')

mask = df['TotalCharges'].isna()
df.loc[mask, 'TotalCharges'] = df.loc[mask, 'MonthlyCharges']

df['SeniorCitizen'] = df['SeniorCitizen'].map({0: 'No', 1: 'Yes'})
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})

print(f'Remaining missing values: {df.drop(columns="customerID").isna().sum().
  ↪sum()}')
```

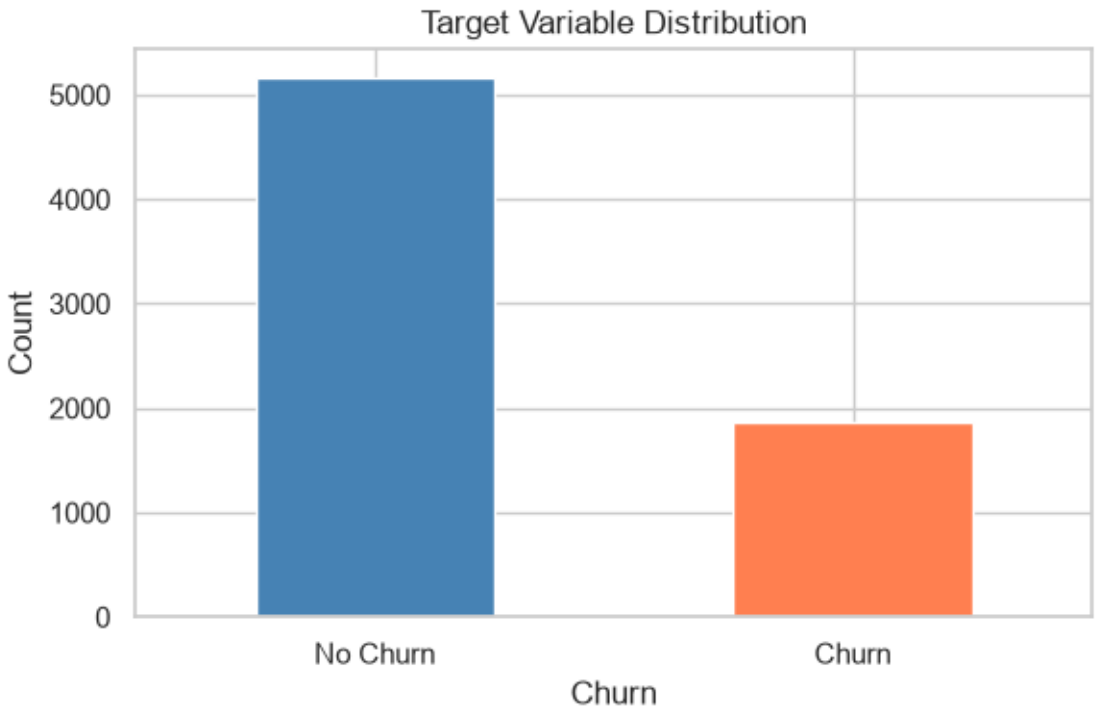
```
Blank TotalCharges values: 11
TotalCharges missing after conversion: 11
Remaining missing values: 0
```

2.0.4 Exploratory Data Analysis

[5]:
```
churn_rate = df['Churn'].value_counts(normalize=True)
print(f'Churn distribution:\n  No churn: {churn_rate[0]:.1%}\n  Churn:    ␣
  ↪{churn_rate[1]:.1%}')

fig, ax = plt.subplots(figsize=(6, 4))
df['Churn'].value_counts().plot(kind='bar', color=['steelblue', 'coral'], ax=ax)
ax.set_xticklabels(['No Churn', 'Churn'], rotation=0)
ax.set_title('Target Variable Distribution')
ax.set_ylabel('Count')
plt.tight_layout()
plt.show()
```

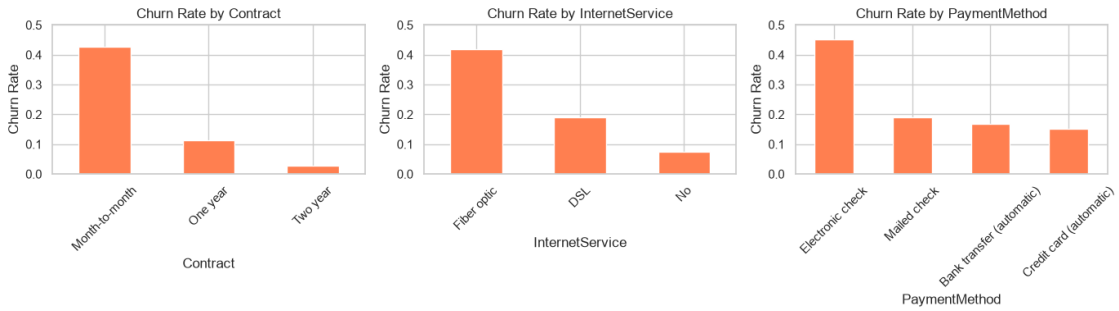
```
Churn distribution:
  No churn: 73.5%
  Churn:    26.5%
```



[6]:
```
fig, axes = plt.subplots(1, 3, figsize=(14, 4))

for ax, col in zip(axes, ['Contract', 'InternetService', 'PaymentMethod']):
    churn_by = df.groupby(col)['Churn'].mean().sort_values(ascending=False)
    churn_by.plot(kind='bar', ax=ax, color='coral')
    ax.set_title(f'Churn Rate by {col}')
    ax.set_ylabel('Churn Rate')
    ax.set_ylim(0, 0.5)
    ax.tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```

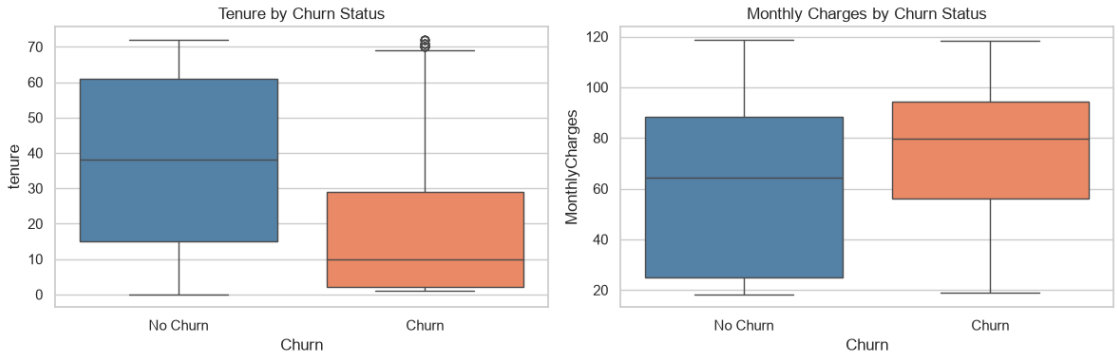


[7]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

sns.boxplot(data=df, x='Churn', y='tenure', ax=axes[0], palette=['steelblue',␣
 ↪'coral'])
axes[0].set_xticklabels(['No Churn', 'Churn'])
axes[0].set_title('Tenure by Churn Status')

sns.boxplot(data=df, x='Churn', y='MonthlyCharges', ax=axes[1],␣
 ↪palette=['steelblue', 'coral'])
axes[1].set_xticklabels(['No Churn', 'Churn'])
axes[1].set_title('Monthly Charges by Churn Status')

plt.tight_layout()
plt.show()
```

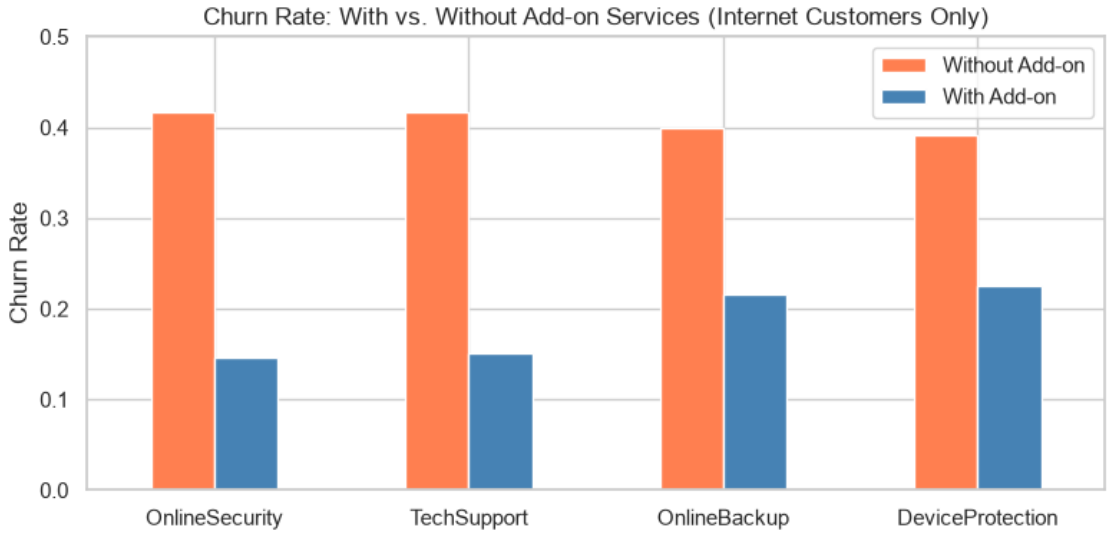


[8]:
```
addon_cols = ['OnlineSecurity', 'TechSupport', 'OnlineBackup',␣
 ↪'DeviceProtection']
addon_churn = {}
for col in addon_cols:
    addon_churn[col] = df[df['InternetService'] != 'No'].groupby(col)['Churn'].
 ↪mean()

addon_df = pd.DataFrame(addon_churn).T
addon_df = addon_df[['No', 'Yes']].rename(columns={'No': 'Without Add-on',␣
 ↪'Yes': 'With Add-on'})

addon_df.plot(kind='bar', figsize=(8, 4), color=['coral', 'steelblue'])
plt.title('Churn Rate: With vs. Without Add-on Services (Internet Customers␣
 ↪Only)')
plt.ylabel('Churn Rate')
plt.xticks(rotation=0)
plt.ylim(0, 0.5)
```

```
plt.legend(loc='upper right')
plt.tight_layout()
plt.show()
```



2.0.5 Data Preparation for Modeling

[9]:
```
model_df = df.drop(columns=['customerID'])

y = model_df['Churn']
X = model_df.drop(columns=['Churn'])

cat_cols = X.select_dtypes(include='object').columns.tolist()
num_cols = X.select_dtypes(include='number').columns.tolist()

print(f'Categorical features ({len(cat_cols)}): {cat_cols}')
print(f'Numeric features ({len(num_cols)}): {num_cols}')
print(f'\nTarget distribution:\n{y.value_counts(normalize=True).to_string()}')

X_encoded = pd.get_dummies(X, columns=cat_cols, drop_first=True, dtype=int)
print(f'\nFeatures after encoding: {X_encoded.shape[1]}')

X_train, X_test, y_train, y_test = train_test_split(
    X_encoded, y, test_size=0.20, random_state=42, stratify=y
)
print(f'\nTraining set: {X_train.shape[0]} rows')
print(f'Test set:     {X_test.shape[0]} rows')
print(f'Train churn rate: {y_train.mean():.3f}')
print(f'Test churn rate:  {y_test.mean():.3f}')
```

```
scaler = StandardScaler()
X_train_scaled = X_train.copy()
X_test_scaled = X_test.copy()
X_train_scaled[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test_scaled[num_cols] = scaler.transform(X_test[num_cols])
```

```
Categorical features (16): ['gender', 'SeniorCitizen', 'Partner', 'Dependents',
'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod']
Numeric features (3): ['tenure', 'MonthlyCharges', 'TotalCharges']

Target distribution:
Churn
0    0.73463
1    0.26537

Features after encoding: 30

Training set: 5634 rows
Test set:     1409 rows
Train churn rate: 0.265
Test churn rate:  0.265
```

2.0.6 Model Training and Evaluation

[10]:
```
def evaluate_model(name, model, X_tr, X_te, y_tr, y_te):
    """Train a model, predict on the test set, and return key metrics."""
    model.fit(X_tr, y_tr)
    y_pred = model.predict(X_te)
    y_prob = model.predict_proba(X_te)[:, 1]

    acc = accuracy_score(y_te, y_pred)
    auc = roc_auc_score(y_te, y_prob)
    report = classification_report(y_te, y_pred, output_dict=True)

    print(f'===== {name} =====')
    print(f'Accuracy:  {acc:.4f}')
    print(f'AUC-ROC:   {auc:.4f}')
    print(f'Precision: {report["1"]["precision"]:.4f}')
    print(f'Recall:    {report["1"]["recall"]:.4f}')
    print(f'F1-score:  {report["1"]["f1-score"]:.4f}')
    print()
    print(classification_report(y_te, y_pred, target_names=['No Churn',␣
 ↪'Churn']))
```

```
    return {
        'Model': name,
        'Accuracy': acc,
        'Precision': report['1']['precision'],
        'Recall': report['1']['recall'],
        'F1-score': report['1']['f1-score'],
        'AUC-ROC': auc,
        'y_prob': y_prob
    }
```

[11]:
```
baseline_model = DummyClassifier(strategy='most_frequent', random_state=42)
baseline_results = evaluate_model('Naive Baseline', baseline_model,
                                  X_train_scaled, X_test_scaled, y_train,␣
 ↪y_test)
```

```
===== Naive Baseline =====
Accuracy:  0.7346
AUC-ROC:   0.5000
Precision: 0.0000
Recall:    0.0000
F1-score:  0.0000

              precision    recall  f1-score   support

    No Churn       0.73      1.00      0.85      1035
       Churn       0.00      0.00      0.00       374

    accuracy                           0.73      1409
   macro avg       0.37      0.50      0.42      1409
weighted avg       0.54      0.73      0.62      1409
```

[12]:
```
lr_model = LogisticRegression(max_iter=1000, class_weight='balanced',␣
 ↪random_state=42)
lr_results = evaluate_model('Logistic Regression', lr_model,
                            X_train_scaled, X_test_scaled, y_train, y_test)
```

```
===== Logistic Regression =====
Accuracy:  0.7381
AUC-ROC:   0.8418
Precision: 0.5043
Recall:    0.7834
F1-score:  0.6136

              precision    recall  f1-score   support

    No Churn       0.90      0.72      0.80      1035
       Churn       0.50      0.78      0.61       374
```

```
    accuracy                           0.74      1409
   macro avg       0.70      0.75      0.71      1409
weighted avg       0.80      0.74      0.75      1409
```

[13]:
```
dt_model = DecisionTreeClassifier(max_depth=5, class_weight='balanced',␣
 ↪random_state=42)
dt_results = evaluate_model('Decision Tree', dt_model,
                            X_train_scaled, X_test_scaled, y_train, y_test)
```

```
===== Decision Tree =====
Accuracy:  0.7346
AUC-ROC:   0.8308
Precision: 0.5000
Recall:    0.8075
F1-score:  0.6176

              precision    recall  f1-score   support

    No Churn       0.91      0.71      0.80      1035
       Churn       0.50      0.81      0.62       374

    accuracy                           0.73      1409
   macro avg       0.71      0.76      0.71      1409
weighted avg       0.80      0.73      0.75      1409
```

[14]:
```
rf_model = RandomForestClassifier(n_estimators=200, max_depth=10,
                                  class_weight='balanced', random_state=42)
rf_results = evaluate_model('Random Forest', rf_model,
                            X_train_scaled, X_test_scaled, y_train, y_test)
```

```
===== Random Forest =====
Accuracy:  0.7566
AUC-ROC:   0.8409
Precision: 0.5281
Recall:    0.7781
F1-score:  0.6292

              precision    recall  f1-score   support

    No Churn       0.90      0.75      0.82      1035
       Churn       0.53      0.78      0.63       374

    accuracy                           0.76      1409
   macro avg       0.72      0.76      0.72      1409
weighted avg       0.80      0.76      0.77      1409
```

2.0.7 Model Comparison Summary

[15]:
```
all_results = [baseline_results, lr_results, dt_results, rf_results]
results_df = pd.DataFrame(all_results).drop(columns='y_prob')
results_df = results_df.set_index('Model')

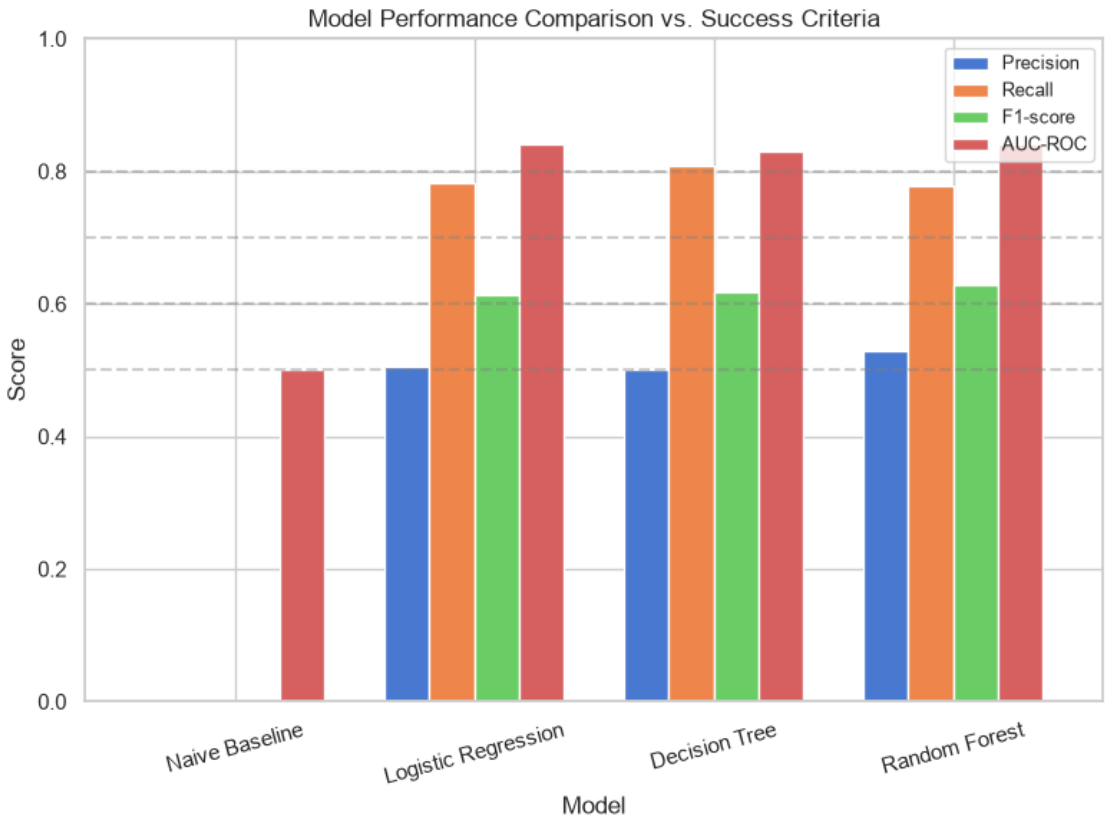
success_criteria = {
    'Recall': 0.70, 'AUC-ROC': 0.80, 'Precision': 0.50, 'F1-score': 0.60
}

print('Model Comparison (test set)')
print(results_df.round(4).to_string())
print()
print('Success Criteria:')
for metric, threshold in success_criteria.items():
    print(f'  {metric}: >= {threshold}')
```

```
Model Comparison (test set)
                     Accuracy  Precision  Recall  F1-score  AUC-ROC
Model
Naive Baseline         0.7346     0.0000  0.0000    0.0000   0.5000
Logistic Regression    0.7381     0.5043  0.7834    0.6136   0.8418
Decision Tree          0.7346     0.5000  0.8075    0.6176   0.8308
Random Forest          0.7566     0.5281  0.7781    0.6292   0.8409

Success Criteria:
  Recall: >= 0.7
  AUC-ROC: >= 0.8
  Precision: >= 0.5
  F1-score: >= 0.6
```

[16]:
```
fig, ax = plt.subplots(figsize=(8, 6))
plot_df = results_df.drop(columns='Accuracy')
plot_df.plot(kind='bar', ax=ax, width=0.75)
for metric, threshold in success_criteria.items():
    ax.axhline(y=threshold, linestyle='--', alpha=0.4, color='gray')
ax.set_title('Model Performance Comparison vs. Success Criteria')
ax.set_ylabel('Score')
ax.set_ylim(0, 1.0)
ax.tick_params(axis='x', rotation=15)
ax.legend(loc='upper right', fontsize=9)
plt.tight_layout()
plt.show()
```

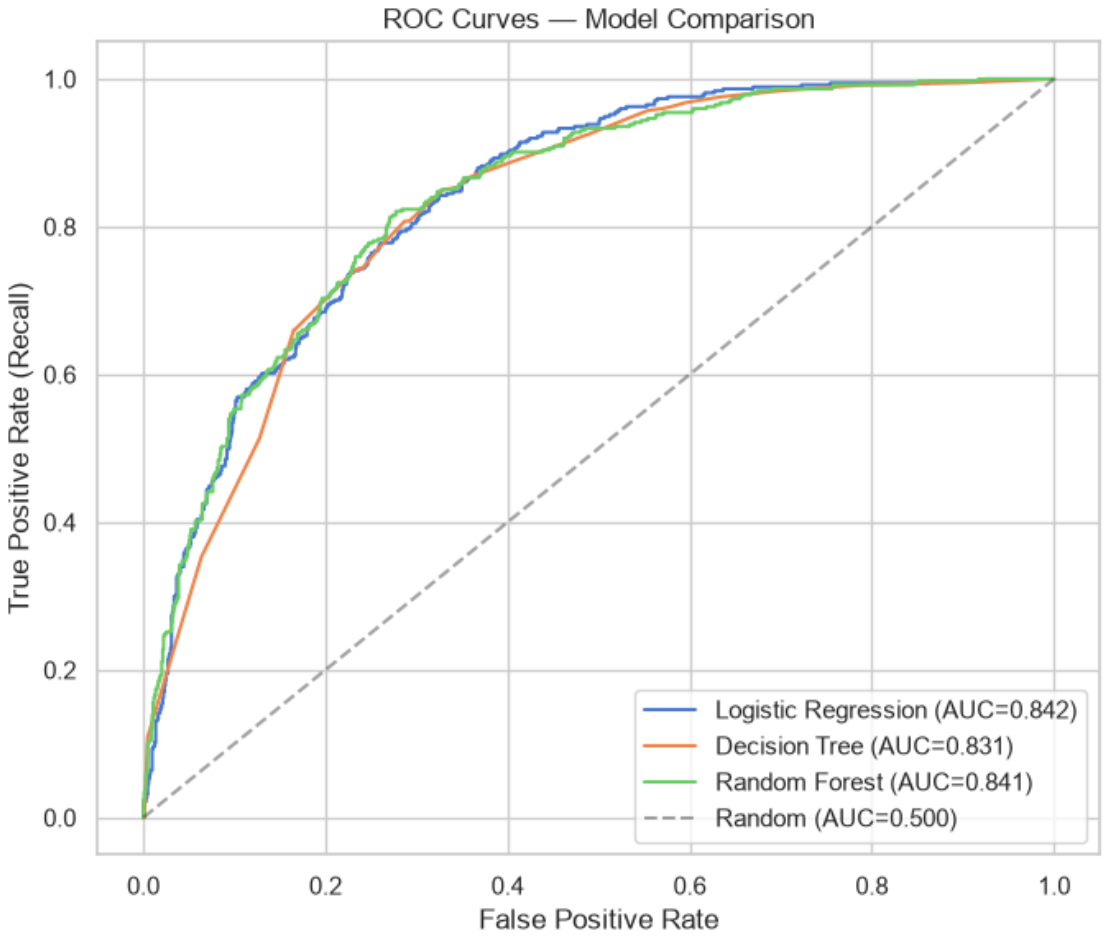


2.0.8 ROC Curves

[17]:
```
fig, ax = plt.subplots(figsize=(7, 6))

for result in all_results[1:]:
    fpr, tpr, _ = roc_curve(y_test, result['y_prob'])
    ax.plot(fpr, tpr, label=f"{result['Model']} (AUC={result['AUC-ROC']:.3f})")

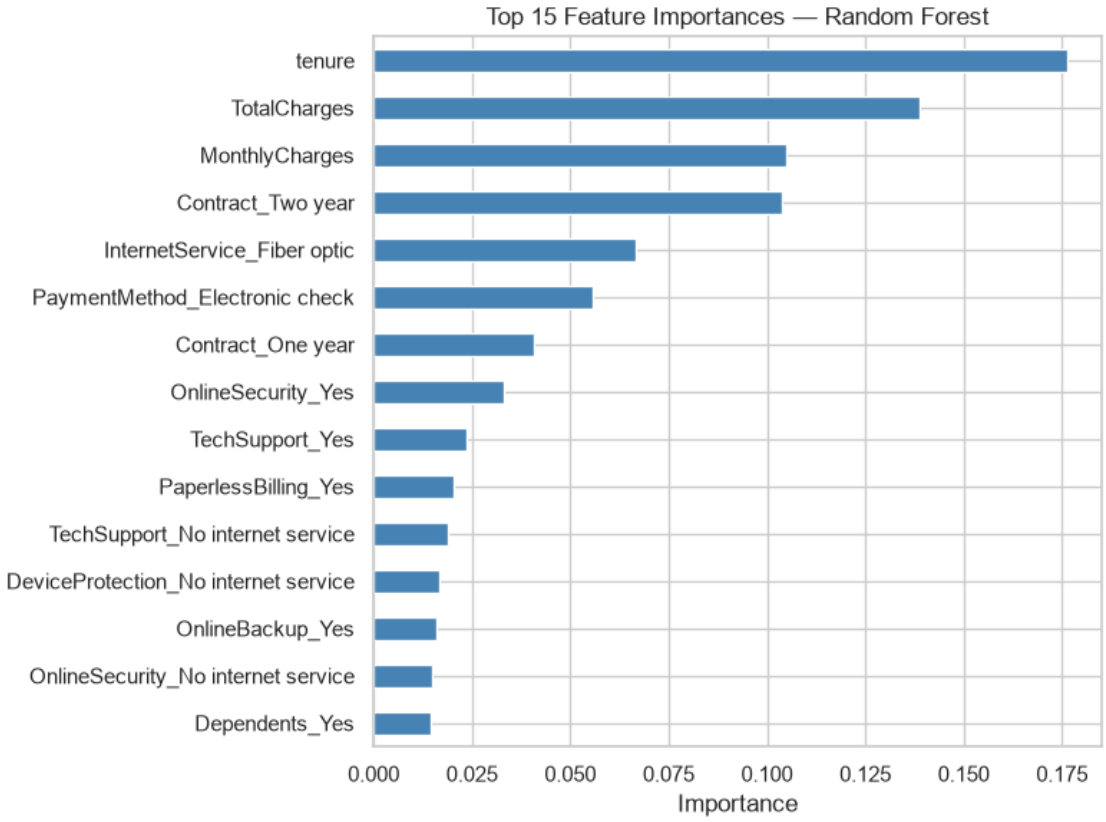
ax.plot([0, 1], [0, 1], 'k--', alpha=0.4, label='Random (AUC=0.500)')
ax.set_xlabel('False Positive Rate')
ax.set_ylabel('True Positive Rate (Recall)')
ax.set_title('ROC Curves - Model Comparison')
ax.legend(loc='lower right')
plt.tight_layout()
plt.show()
```



2.0.9 Feature Importance (Random Forest)

[18]:
```
importances = pd.Series(rf_model.feature_importances_, index=X_encoded.columns)
top_features = importances.nlargest(15)

fig, ax = plt.subplots(figsize=(8, 6))
top_features.sort_values().plot(kind='barh', ax=ax, color='steelblue')
ax.set_title('Top 15 Feature Importances - Random Forest')
ax.set_xlabel('Importance')
plt.tight_layout()
plt.show()
```



2.0.10 Logistic Regression Coefficients

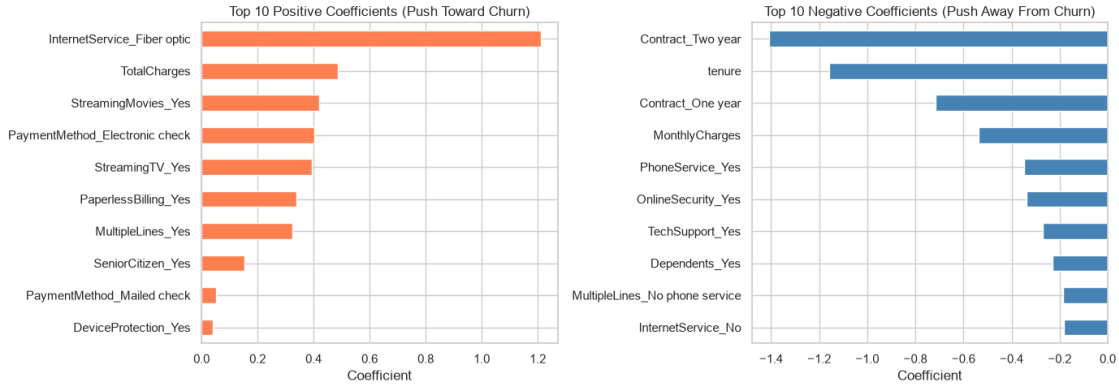
[19]:
```
coefs = pd.Series(lr_model.coef_[0], index=X_encoded.columns)
top_positive = coefs.nlargest(10)
top_negative = coefs.nsmallest(10)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

top_positive.sort_values().plot(kind='barh', ax=axes[0], color='coral')
axes[0].set_title('Top 10 Positive Coefficients (Push Toward Churn)')
axes[0].set_xlabel('Coefficient')

top_negative.sort_values(ascending=False).plot(kind='barh', ax=axes[1],␣
 ↪color='steelblue')
axes[1].set_title('Top 10 Negative Coefficients (Push Away From Churn)')
axes[1].set_xlabel('Coefficient')

plt.tight_layout()
plt.show()
```



2.0.11 Fairness Analysis

[20]:
```
print('=== Fairness Analysis: Random Forest Performance by Demographic Group␣
↪===\n')

rf_pred = rf_model.predict(X_test_scaled)

demo_cols = {
    'gender': 'gender_Male',
    'SeniorCitizen': 'SeniorCitizen_Yes'
}

for label, col in demo_cols.items():
    print(f'--- {label} ---')
    for val in [0, 1]:
        group_mask = X_test_scaled[col] == val
        group_name = f'{label}={"Yes" if val == 1 else "No"}'
        y_true_group = y_test[group_mask]
        y_pred_group = rf_pred[group_mask]
        report = classification_report(y_true_group, y_pred_group,␣
↪output_dict=True, zero_division=0)
        print(f'  {group_name}: Recall={report["1"]["recall"]:.3f}, '
              f'Precision={report["1"]["precision"]:.3f}, '
              f'F1={report["1"]["f1-score"]:.3f}, '
              f'n={group_mask.sum()}')
    print()
```

```
=== Fairness Analysis: Random Forest Performance by Demographic Group ===

--- gender ---
  gender=No: Recall=0.751, Precision=0.541, F1=0.629, n=687
  gender=Yes: Recall=0.807, Precision=0.516, F1=0.629, n=722
```

```
--- SeniorCitizen ---
  SeniorCitizen=No: Recall=0.739, Precision=0.501, F1=0.597, n=1187
  SeniorCitizen=Yes: Recall=0.888, Precision=0.604, F1=0.719, n=222
```

2.0.12 Confusion Matrix — Random Forest (Recommended Model)

[21]:
```
cm = confusion_matrix(y_test, rf_pred)

fig, ax = plt.subplots(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=ax,
            xticklabels=['No Churn', 'Churn'],
            yticklabels=['No Churn', 'Churn'])
ax.set_xlabel('Predicted')
ax.set_ylabel('Actual')
ax.set_title('Confusion Matrix - Random Forest')
plt.tight_layout()
plt.show()
```

